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Strengthening mechanisms in Fe-Al based ferritic low-density steels

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ABSTRACT

Low-density steels with different aluminium contents have been investigated with an aim to examine the occurrence of different strengthening mechanisms leading to its higher strength. A composition corresponding to 6.8 wt% aluminium has been studied to understand the underlying strengthening mechanisms. Different factors contributing to the strengthening mechanisms have been separately analyzed. Microstructural features have been analyzed using Mössbauer spectroscopy, small angle X-ray scattering (SAXS), X-ray line profile analysis and transmission electron microscopy (TEM). The enhanced yield strength of the low-density steel containing 6.8 wt % Al was attributed to the strengthening effects arising from the ferrite grain size, dislocations incorporated during processing, ordered phase formation and the presence of Al atoms in the solid solution. Each of these operating mechanisms was modelled by using its constitutive equation for example, grain size strengthening by classical Hall-Petch equation and the strengthening from dislocations by Taylor's equation. In addition, the formation of nano-sized ordered phase was evaluated by TEM, Mössbauer spectroscopy, SAXS and hence order strengthening was modelled by using the size and volume fraction (as determined by TEM and SAXS). Strengthening due to lattice frictional stress required for dislocation motion was also incorporated into the model.

1. Introduction

The reduction in weight of automobile components is essential for decreasing fuel consumption and hence, the exhaust gas emissions [1]. Due to its popular choice in automotive components, the density reduction of steels by the addition of aluminium (Al) has become an attractive proposition to reduce the weight of the vehicle. Addition of aluminium up to ~10 wt% leads to the formation of ferrite (Body-Centered Cubic (BCC)) phase [2] along with the density reduction of 8% [3,4]. These steels, depending on the alloying addition, can form a single phase (ferrite) [3], dual phase (ferrite + austenite) [5] or multiphase structures (ferrite + austenite + κ-carbides) [4,6,7].

Rana et al. [3] reported the development of a single phase binary Fe-Al alloy with the density reduction up to 12.7% upon the addition of 9.7 wt% aluminium. Upon the addition of Al, a reduction in Young's modulus was also reported in Ref. [3] therefore limiting its use in automotive components. The same authors in Refs. [8,9] also stabilized a duplex microstructure (ferrite + austenite) by the addition of low amounts of manganese (Mn = 3.34 wt%) and carbon (C = 0.18 wt%), which led to the precipitation of κ-carbides. The produced lean alloyed Fe-Al-Mn-C steel did not provide much promising mechanical properties (YS = 472 MPa, UTS = 635 MPa, elongation = 19.7%) in comparison to binary Fe-Al alloys. Excessive addition of manganese (> 25 wt%) along with aluminium and carbon by several other authors [10–12] have produced a multiphase microstructure producing better mechanical properties. In such cases, the precipitation of nano-sized κ-carbides in austenite matrix was responsible for the significant improvement of mechanical properties. However, dislocation motion has been attributed to consist of planar glide in spite of low stacking fault energy, possibly due to shearing of the κ-carbides particles. Steels with Mn content greater than 20 wt% also shows the formation of deformation induced ε and α′ martensite along with deformation twinning. The prevalence of twinning or martensite formation is dependent on the stacking fault energy of the steel [13,14]. However, the addition of high amount of manganese (> 25 wt%) may lead to increased cost of steel production limiting their use in automobile parts. Low-density steels having a duplex microstructure containing ferrite and austenite has also been reported [13]. These steels contain Mn (3–6 wt%) and Al (3–6 wt%) along with C (0.1–0.4 wt%). These steels deform by strain accommodation by ferrite along with deformation induced martensite formation from austenite. The substructure after deformation consisted of planar glide in austenite along with wavy slip in ferrite.

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The binary Fe-Al based low-density steels compositions developed by Rana et al. [3] was reported to possess moderate strength and ductility combination. Thus, novel strategies are needed to improve the strength of binary Fe-Al alloy with lower cost to suit for its application in automobile bodies, which requires a comprehensive understanding of the role of aluminium in the strengthening mechanisms. This paper elucidates the study of high strength low-density steels based on the binary Fe-Al composition, which could provide the basis of alloy development in this direction. A rather fine-scale characterization using transmission electron microscopy (TEM), Mössbauer spectroscopy and small angle X-ray scattering (SAXS) have been carried out to examine the formation of ordered phases, while the microstructural features such as dislocation density have been analyzed using X-ray line profile analysis.

2. Experimental procedure and methods of analyses

The charges consisting of interstitial free (IF) steel and pure aluminium were melted in a vacuum arc furnace to prepare low-density steels with aluminium contents of 5, 6, 6.8 and 9 wt%. The obtained circular buttons of diameter 10 cm and height 5 cm were re-melted multiple times (7–8 times) to remove segregation. The compositions after heat treatment were examined by wavelength dispersive spectroscopy (WDS) in a JEOL JXA-8530F electron probe micro-analyzer (EPMA) operating at an accelerating voltage of 15 kV. The determined compositions have been reported in Ref. [15]. Density measurements were undertaken by applying Archimedes principle [15]. The composition of the material throughout the text corresponds to the aluminium content in wt%.

Samples for Mössbauer spectroscopy and SAXS were prepared by filing to obtain powders. The powder samples were then wrapped in a copper foil and sealed in a glass tube under high vacuum (10^{-5} mBar) and annealing was performed at 600 °C for 60 min. The samples were annealed to remove the internal stress generated during the creation of powder particles by filing. Mössbauer spectroscopy was carried out at room temperature in transmission geometry by using a conventional constant acceleration Doppler velocity drive with ^{57}Co (Rh-matrix) source. The velocity of the drive was calibrated by a pure iron (α-Fe) foil.

SAXS measurements were carried out by using a laboratory-based dual energy Mo-Cr-SAXS system (Xeuss, Xenocs, France) [16]. Use of dual X-ray energies and different camera lengths of 550 and 2400 mm allows covering of a wide q-range from 0.0254 to 10 nm^{-1}, where q $\left(=\frac{4\pi}{\lambda}\sin\theta\right)$ is the modulus of momentum transfer. The parameters λ and θ are the wavelength of the incident X-ray and half of the scattering angle, respectively. Samples for SAXS measurements were prepared by sandwiching a thin layer of the powdered material between Scotch Magic tape (3 M, USA). 2D-SAXS profiles were measured in transmission geometry with a measurement time of 4–6 h. The 2D-SAXS profiles were then azimuthally integrated to 1D profiles using FIT2D software [17]. Further, the SAXS intensity profiles, I(q), were converted into absolute scale with the scale factor obtained using glassy carbon as a secondary standard [18], after performing standard subtraction of background, noise and scattering due to the scotch tape. Silver-Behenate powder sample was used to calibrate the scattering angle [19].

The Fe-6.8Al alloy was cold rolled (CR) to 98% reduction in thickness and annealed at 750 °C for 5 min and 60 min. X-Ray Diffraction (XRD) measurements were performed using X-Pert PRO diffractometer with Cu-Kα radiation. All the samples were scanned from an angle of 40° to 120° with a step size of 0.03°. For the determination of dislocation density, only the (110) XRD peak profile was used in the present study. The dislocation density was determined using the Variance method developed by Wilson and Groma [20–23]. Microstructural investigation of the Fe-6.8Al sample in cold rolled and annealed conditions were carried out using FEI-ESEM Quanta 200 scanning electron microscope (SEM) operating at an accelerating voltage of 25 kV. Sub size tensile specimens were prepared with a gauge length of 6 mm and width of 2 mm as reported previously [15]. The gauge section was machined parallel to rolling direction (RD) of the rolled sheet. Three tensile tests were conducted for each sample condition. Uniaxial tensile tests were performed at a strain rate of $10^{-3}\,s^{-1}$ in an Instron 5967 screw driven tensile testing machine. The presence of ordered phase was also detected by TEM investigation of Fe-6.8Al sample. The sample preparation for TEM was carried out by grinding and twin jet electropolishing using a solution comprising of 78% Methanol, 10% lactic acid, 7% sulfuric acid, 3% nitric acid and 2% hydrochloric acid.

2.1. Analysis of Mössbauer spectrum

The Mössbauer spectrum of an alloy can be described by a superimposition of the distribution of sextets corresponding to different values of the hyperfine magnetic field (B_{hf}). So, the experimentally measured Mössbauer spectrum of each Fe-Al composition was fitted by the NORMOS program to obtain the probability distribution ($P(B_{hf})$) of the hyperfine magnetic field, B_{hf} [2]. The B_{hf} distribution consists of up to 40 sextets. The observed B_{hf} is related to the local atomic configuration comprising of first and second nearest non-magnetic neighbours by the equation [24,25]

$$B = B_o - n_1\Delta B_1 - n_2\Delta B_2 \quad (1)$$

where B_o = 33 T, ΔB_1 = 2.4 T and ΔB_2 = 0.5 T. In the present investigation, the values of ΔB_1 and ΔB_2 were obtained from Ref. [26]. In Eq. (1), n_1 and n_2 are the number of aluminium atoms in the first and second nearest neighbours.

2.2. Analysis of SAXS profiles

The size, volume fraction and number density of an ordered DO_3 phase in the material can be quantitatively estimated from the SAXS intensity profiles, I(q) with a local mono-disperse model using non-linear curve fitting analysis. According to this model, the scattering intensity I(q) is described by [27]

$$I(q) = \Delta\rho^2 d_N \int_0^R N(r)[V(r)F(q, r)]^2 dr \quad (2)$$

where Δρ is the scattering length difference between Fe_3Al (DO_3) ordered phase and disordered Fe-Al matrix, d_N and V(r) are the number density and volume of the ordered phase respectively. N(r) is the normalized number density distribution function with the radius of ordered phases (r), where it is assumed that the distribution of the ordered phase is following log-normal distribution and is given by [27]

$$N(r) = \frac{1}{rw\sqrt{2\pi}}\exp\left[-\frac{(\ln(r) - \ln(r_p))^2}{2w^2}\right] \quad (3)$$

In Eq. (3) r_p and w are the peak radius of the ordered phase and standard deviation of log-normal distribution, respectively. Since the area under the N(r) distribution equals unity, the real number density distribution is expressed as a product of d_N and N(D) with D (= 2r). The average size of the ordered phase is calculated as follows [27]

$$D_{ave} = 2r_p \exp\left(\frac{w^2}{2}\right) \quad (4)$$

F(q, r) in Eq. (2) is the form factor of the ordered phase, which is assumed to be spherical. The form factor for spherical DO_3 ordered phase modelled by SAXS is described as [28]

$$F(q, r) = \frac{3[\sin(qr) - (qr)\cos(qr)]}{(qr)^3} \quad (5)$$

when q approaches zero, $F^2(q,r)$ equates to 1, whereas when q approaches a value larger than r, an asymptotic behaviour of $F^2(q,r)$ is observed and has a q^{-4} dependence with q, which is known as Porod scattering [29].

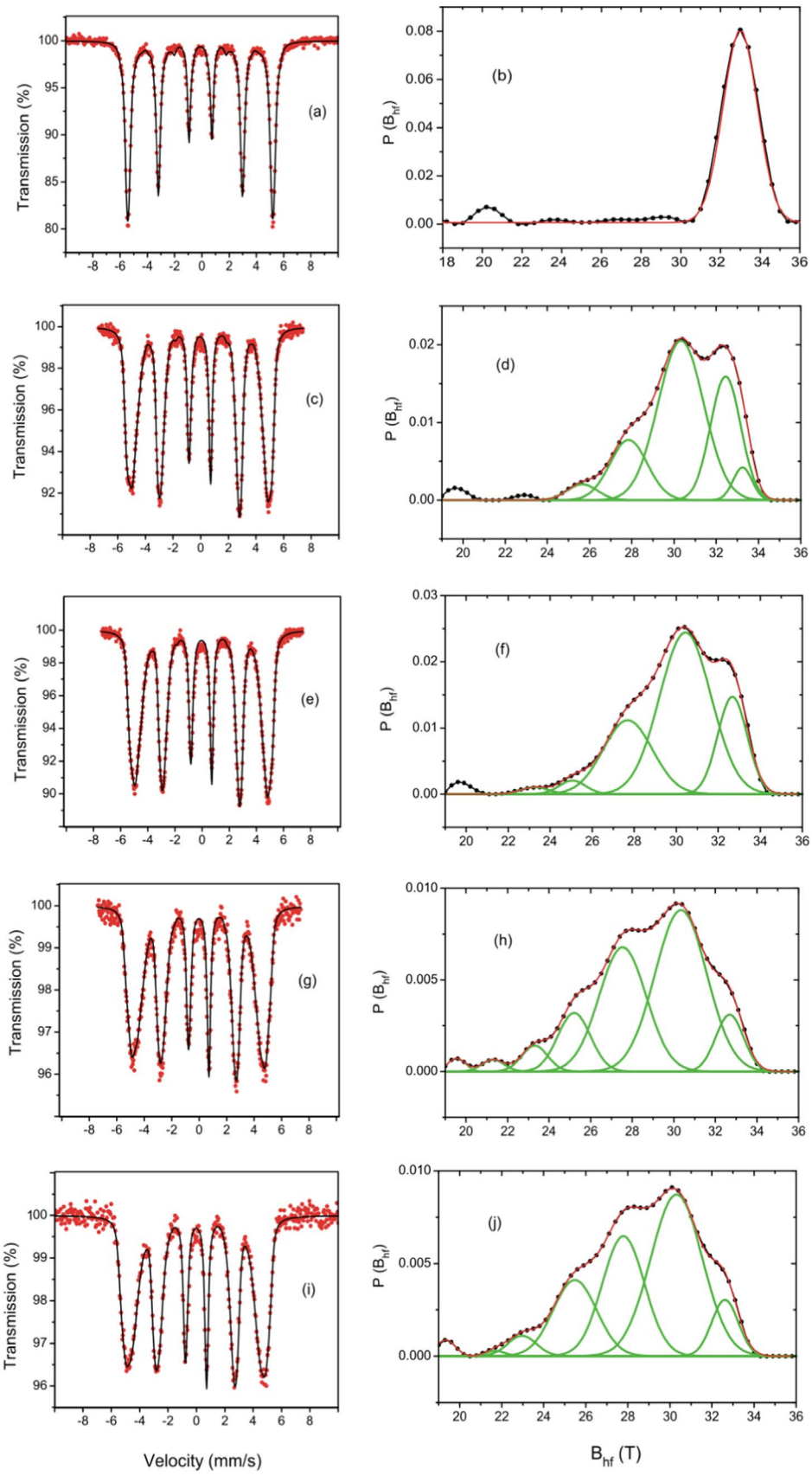


Fig. 1. The Mössbauer spectra (left) and the corresponding B_{hf} distribution ($P(B_{hf})$) (right) for (a, b) α-Fe-foil, (c, d) Fe-5Al, (e, f) Fe-6Al, (g, h) Fe-6.8Al and (i, j) Fe-9Al low-density steels. The humps in $P(B_{hf})$ are deconvoulted.

Table 1
Hyperfine magnetic field (B_{hf}), Spectral area, Volume fraction (V_f) and proposed assignments of the atomic configurations that produce the respective magnetic hyperfine field.

Composition	B_{hf} (T) (± 0.5 T)	Area (%) (± 1%)	V_f (%) (± 1%)	Assignment (atomic configurations)
α-Fe-foil	33	100	100	(8Fe)nn + (6Fe)nnn
Fe-5Al	32.67	24	23.55	(8Fe)nn + (5Fe + 1Al)nnn
	30.65	51.6	50.7	(7Fe + 1Al) nn + (6Fe)nnn Or (8Fe)nn + (6Al)nnn
	28.22	23.75	25	(6Fe + 2Al)nn + (6Fe)nnn
	25.39	0.65	1	(5Fe + 3Al)nn + (6Fe)nnn
Fe-6Al	32. 67	18.6	17.6	(8Fe)nn + (5Fe + 1Al)nnn
	30.44	54.16	53.5	(7Fe + 1Al) nn + (6Fe)nnn Or (8Fe)nn + (6Al)nnn
	27.69	23.63	24.8	(6Fe + 2Al)nn + (6Fe)nnn
	25.05	2.35	2.6	(5Fe + 3Al)nn + (6Fe)nnn
	23.26	1.26	1.4	(4Fe + 4Al)nn + (6Fe)nnn
Fe-6.8Al	32.68	8.55	7.7	(8Fe)nn + (5Fe + 1Al)nnn
	30.34	45.45	44	(7Fe + 1Al) nn + (6Fe)nnn Or (8Fe)nn + (6Al)nnn
	27.54	31.9	33	(6Fe + 2Al)nn + (4Fe + 2Al) nnn
	25.21	10.4	11.4	(5Fe + 3Al)nn + (5Fe + 1Al) nnn
	23.31	3.7	4.2	(4Fe + 4Al)nn + (6Fe)nnn
Fe-9Al	32.62	7.9	7.2	(8Fe)nn + (5Fe + 1Al)nnn
	30.31	44.1	42.3	(7Fe + 1Al) nn + (6Fe)nnn Or (8Fe)nn + (6Al)nnn
	27.79	26.8	27.8	(6Fe + 2Al)nn + (6Fe)nnn
	25.48	17.5	18.9	(5Fe + 3Al)nn + (6Fe)nnn
	22.97	3.1	3.6	(4Fe + 4Al)nn + (6Fe)nnn

2.3. Analysis of X-Ray diffraction pattern

The dislocation density of disordered Fe-Al matrix (ferrite) was determined by analyzing the different order of moments of (110) peak intensity profile. It has been reported by Wilson [21,22,30] and Groma [20,31] that crystallite size and lattice strain create different asymptotic behaviour in the kth order restricted moment. The kth order moment for a peak profile is calculated by using the equation [32,33]

$$M_k(q') = \frac{\int_{-q'}^{q'} q^k I(q)dq}{\int_{-\infty}^{\infty} I(q)dq} \quad (6)$$

where $q = \frac{2}{\lambda}(\sin\theta - \sin\theta_c)$ and I(q) is the intensity distribution of the peak as a function of q. λ is the wavelength of X-ray, θ is the diffraction angle and θ_c is the Bragg angle [32,33]. The asymptotic forms of the 2nd and 4th order restricted moments can be correlated to the microstructural parameters of ferrite matrix such as crystallite size (D_s) and dislocation density (ρ) by the equations [32]

$$M_2(q) = \frac{1}{\pi^2 D_s} q - \frac{L}{4\pi^2 K^2 D_s^2} + \frac{\Lambda <\rho> ln\left(\frac{q}{q_0}\right)}{2\pi^2} \quad (7)$$

and

$$\frac{M_4(q)}{q^2} = \frac{1}{3\pi^2 D_s} q + \frac{\Lambda <\rho>}{4\pi^2} + \frac{3\Lambda^2 <\rho^2> ln\left(\frac{q}{q_1}\right)}{4\pi^2 q^2} \quad (8)$$

where L is the taper parameter which depends on the rate of decrease in the cross-sectional area of the crystallites, K is the Scherrer constant, D_s is the average length of the column or the crystallite size recorded in the direction of the diffracting vector. $<\rho>$ is the average dislocation density, $<\rho^2>$ is the average of the square of the dislocation density, q_0 and q_1 are suitable fitting parameters [32,33]. Λ is the dislocation contrast factor which is a geometric constant and is calculated by using the anisotropic elastic modulus [34–36]. Its value depends on the dislocation slip system present in the sample. For the present study, the dislocation contrast factor has been determined by the program ANIZC [31]. The first two terms of $M_2(q)$ and the first term of $M_4(q)/q^2$ (Eqs. 7 and 8) represents the presence of finite size of the domains in the material [33]. The third term of $M_2(q)$ and the second term from the $M_4(q)/q^2$ (Eqs. 7 and 8) represents the contribution from the dislocation density [32]. Finally, the last term of $M_4(q)/q^2$ represents the contribution from the average of the square of the dislocation density. It is reported in Ref. [33] that if the broadening due to the domain size is negligible which may be due to the presence of a large domain size, then the dislocation density calculated from the second order (Eq. (7)) and fourth order (Eq. (8)) restricted moment are in good agreement. But if the domain size is very small, the dislocation density values evaluated from the fourth order (Eq. (8)) restricted moment are more accurate than that calculated from the second order moment [33]. For selecting a common background for the evaluation of the domain size we have evaluated both the moments as per procedure in Ref. [37].

3. Results

3.1. Microstructural characterization

3.1.1. Mössbauer spectroscopy

Fig. 1 shows the Mössbauer spectra and the corresponding B_{hf} distribution ($P(B_{hf})$) of pure Fe and Fe-Al low-density steels. The Mössbauer spectrum for pure Fe is shown in Fig. 1a. The corresponding B_{hf} distribution is presented in Fig. 1b shows peaks at 33 T and ~20 T for Fe foil. This hyperfine field of 33 T corresponds to a local atomic arrangement of eight Fe atoms as first nearest neighbour (nn) and six Fe atoms as second nearest neighbour (nnn). Such a configuration corresponds to bcc-Fe. The Mössbauer spectra of the Fe-Al low density steels are shown in Fig. 1c,e,g,i. With the addition of Al atoms the appearance of new peaks in the B_{hf} distribution is observed (Fig. 1d,f,h,j). These peaks emerge due to the change in local atomic configurations by addition of non-magnetic Al atoms. Furthermore, the Al atoms in the Fe-Al low-density steels can be present in the lattice either in the form of solid solution or as ordered phase. By using Eq. (1) the local hyperfine field has been calculated for all the samples which is related to the local atomic configuration, as shown in Table 1. Fig. 1c,d shows the Mössbauer spectrum of Fe-5Al steel and the corresponding B_{hf} distribution, respectively. The humps in B_{hf} curve was deconvoluted into multiple peaks, with each peak being associated with a particular field (Table 1). It can be observed that as the number of Al atoms in the first and second nearest neighbour increased, the corresponding hyperfine field decreased which is due to the non-magnetic nature of the Al atom. Similar broad humps were also observed in the B_{hf} distribution of other Fe-xAl low-density steels (x = 6, 6.8, 9) (Fig. 1f,h,j). The calculated hyperfine field values for the Mössbauer spectra of DO_3 (Fe_3Al) and B2 (FeAl) phases in Fe-Al alloys (with Al > 24 wt%) are reported to be 30 T and 21 T respectively [24,25]. We have observed a peak at B_{hf} = 30.6 T in all the studied Fe-Al low-density steel samples. Although the 30.6 T component may be associated with the local atomic configuration of (7Fe + 1Al)nn + (6Fe)nnn, the presence of Fe_3Al phase corresponding to the atomic configuration of (8Fe)nn + (6Al)nnn cannot be ruled out due to the close proximity of the peaks corresponding to the two different configurations. Fe_3Al phase has another associated hyperfine field component ((4Fe + 4Al)nn + (6Fe)nnn) at ~21 T [24]. For the samples with Al content of 6, 6.8 and 9, we have observed low-intensity peaks at ~23 T. This may also be a possible indicator for the presence of Fe_3Al phase with a low volume fraction. The absence of peak at 23 T for the Fe-5Al sample may be due to its low volume fraction.

3.1.2. Small angle X-Ray scattering studies

Fig. 2 shows the 2D SAXS profiles of the Fe-Al alloys. The SAXS

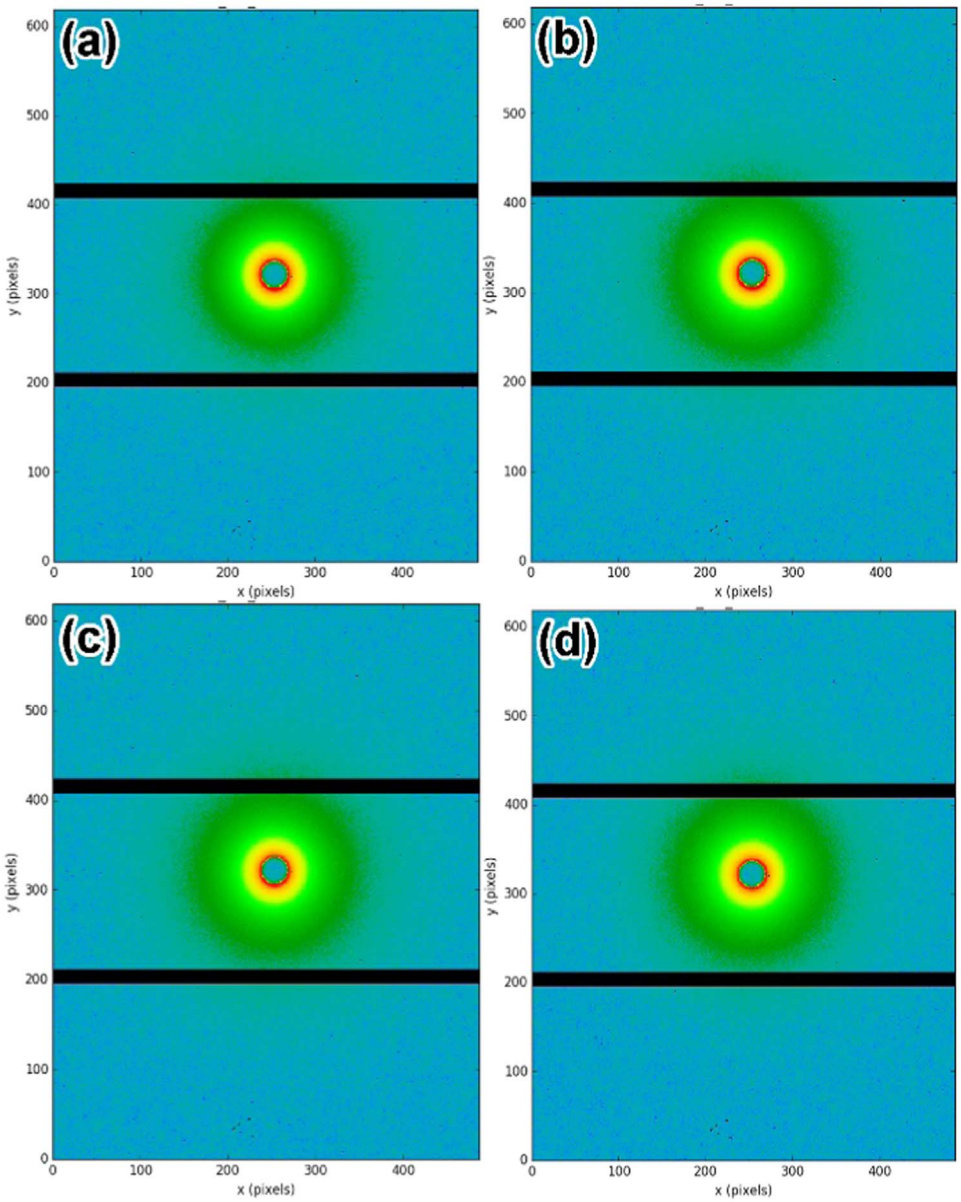


Fig. 2. 2D-SAXS profiles of the (a) Fe-5Al, (b) Fe-6Al, (c) Fe-6.8Al and (d) Fe-9Al low-density steels.

profiles are observed to be radially symmetric. Radial integration has been done and the intensity profiles has been plotted in Fig. 3. Fig. 3 shows the experimental SAXS intensities (open symbols) and fitted curves (solid curves) with increasing Al content from 5 to 9 wt% for Fe-Al low-density steels. The scattering from the nano-sized ordered phase can clearly be observed as a weak peak between q = 0.3–3 nm^{-1}. It can be observed from Fig. 3 that the scattering intensity is increasing with Al-content, implying the increase of nano-sized ordered phase and the Guinier regions are shifting towards the low q-region confirming the increase in sizes of the ordered phase. At high q region (> 3 nm^{-1}), the intensity shows an incoherent flat background, whereas at the low q-region (below 0.3 nm^{-1}), the intensity decays as a function of q^{-4} with q, indicating that this intensity originates from larger heterogeneities that are present in the alloy. The large heterogeneities may be grains and/or grain boundaries of the alloys. This agrees well with various Al-distributed ferrite phase as observed by Mössbauer spectroscopy. The SAXS intensity profiles were fitted with Eq. (3). The fitted curves (solid lines in Fig. 3) agree well with the experimental SAXS intensities. The average diameter and the volume fraction of the ordered phase (Fe_3Al) were estimated from the model curve fitting analysis which is presented in Fig. 4. The diameter of Fe_3Al ordered phase increased from 2.5 ± 0.15 nm to 3.75 ± 0.2 nm with increasing Al content. The increase in volume fraction and size of the ordered phase could be due to the increasing driving force for the order phase formation with increasing Al addition (wt%). It is to be noted that the volume fractions (Fig. 4) of the ordered phase observed by SAXS are lower than that of the DO_3 phase observed by Mössbauer spectroscopy. This is due to the fact that SAXS provides collective information about the ordered phase having a sharper boundary whereas Mössbauer spectroscopy traces all local environments, including those having diffused boundaries.

3.1.3. Electron microscopic investigations

The SEM micrographs of the 98% cold rolled and sample annealed at 750 °C for 5 min has been reported in Ref. [15]. The cold rolled sample (Ref. [15], Fig. 4c) shows heavily elongated and pancaked grains along the rolling directions. The presence of shear bands can be observed indicating non-uniform deformation. Shear bands were observed at an angle of ~40° with the RD. The average width of the deformed lamellae is 2.6 ± 1.2 μm. The sample annealed for 5 min (Ref. [15], Fig. 5c) and 60 min (Fig. 5) shows equiaxed and fully recrystallised grains. The grain/crystallite size has been reported in Table 2. Grain growth can be observed for the sample annealed for

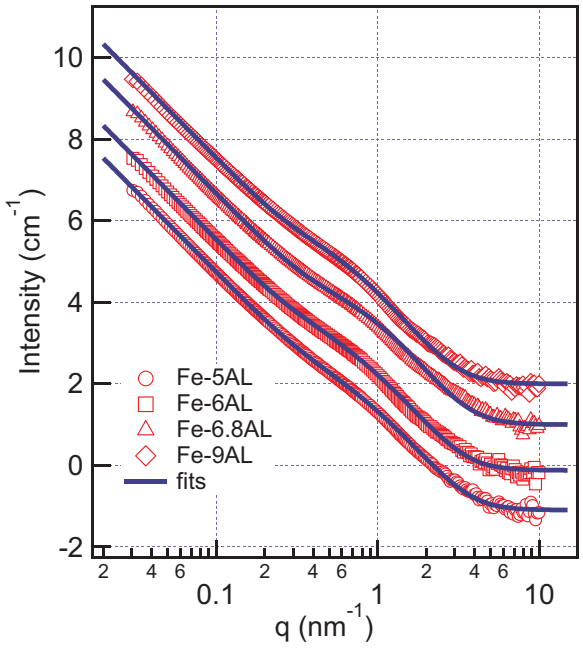


Fig. 3. Variation of logarithmic SAXS intensity as a function of q for Fe-Al low-density steels with different compositions. Open symbols, solid lines are experimental data and fitted curves with Eq. (3), respectively. The curves have been offsetted vertically with a corresponding factor to avoid crowding the plot. The broken lines with dots are proportional to q^{-4}.

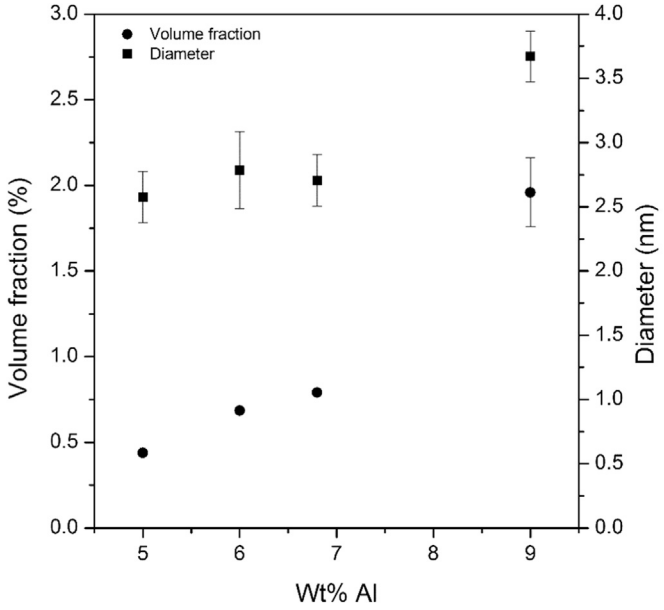


Fig. 4. Variation of diameter and volume fraction of Fe_3Al ordered phase with changing weight percentage of Al.

60 min as compared to 5 min due to increase in grain size (25.2–48.3 μm).

The distribution of ordered phase can also be observed from the dark field TEM micrograph shown in Fig. 6. The diffraction pattern shown in the inset shows the presence of (i) strong and (ii) weak spots (highlighted by yellow arrows). The strong spots are due to reflections from the matrix [011] zone axis whereas the weak spots correspond to superlattice peaks from ordered phase. Thus the diffraction pattern shows the existence of the ordered phase. The dark field image was obtained from the 00weak diffraction spots. The dark field image shows the presence of very fine ordered phase (bright) which are seen to be

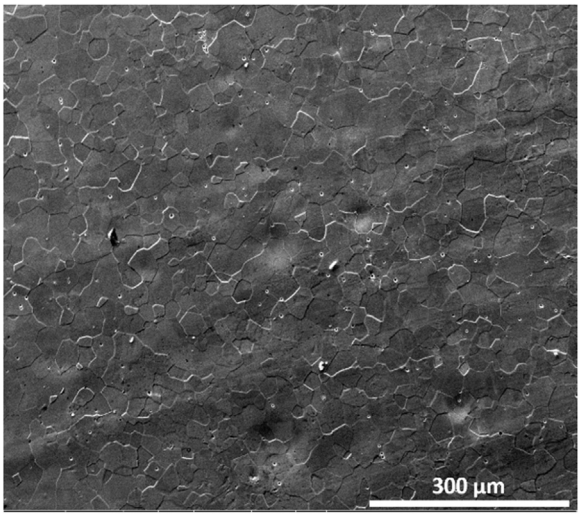


Fig. 5. SEM image of the Fe-6.8Al sample annealed at 750 °C for 60 min.

Table 2
Grain size of the Fe-6.8Al samples in different conditions.

Sample condition	Grain (μm)/crystallite size (nm)	Determination method	Ref.
98% CR	102.3 ± 9.8	XRD analysis	–
750 °C, 5 min	25.2 ± 5.0	SEM	[15]
750 °C, 60 min	48.3 ± 1.2	SEM	–

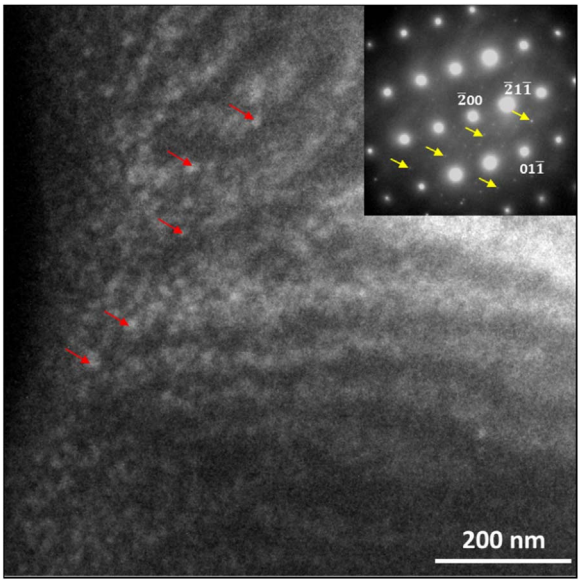


Fig. 6. Dark field TEM micrograph of the Fe-6.8Al sample showing the distribution of ordered phase. The inset shows the diffraction pattern from [011] zone axis of Fe-Al matrix showing weak ordered phase spots (highlighted by yellow arrows). Ordered phase are shown by red arrows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

associated with dark background. The size of the ordered phase has been determined to be 2.1 ± 0.2 nm. The size of the ordered phase measured from TEM is in correlation to that determined from SAXS studies (cf. Fig. 4).

3.1.4. X-Ray diffraction studies

Fig. 7a shows the (110) XRD peak profiles for the Fe-6.8Al samples after 98%CR and annealing at 750 °C for 5 min and 60 min. The full

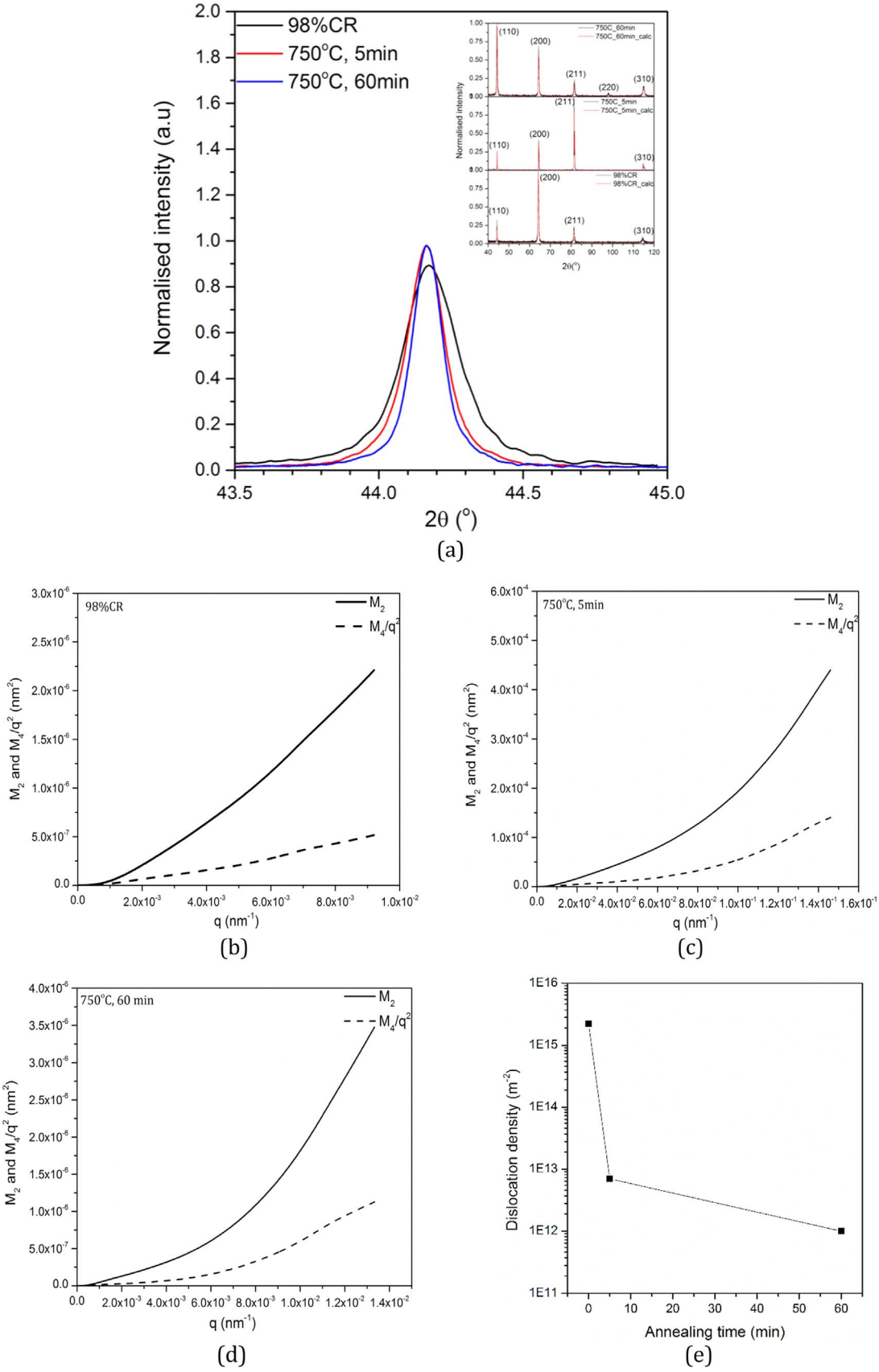


Fig. 7. (a) (110) peak profile, plot of the M_2 and M_4/q^2 with q for the Fe-6.8Al samples after (b) 98% CR, isothermally annealing at 750 °C for (c) 5 min, (d) 60 min and (e) variation of dislocation density for Fe-6.8Al low-density steel in different conditions. Inset in (a) shows the complete XRD profiles.

Table 3
Variation of crystallite size and lattice micro-strain for Fe-6.8Al samples in different conditions.

	Sample condition		
	98% CR	750 °C, 5 min	750 °C, 60 min
Crystallite size (nm)	102.3 ± 9.8	–	–
Micro-strain	1.46×10^{-3}	9.46×10^{-4}	1.46×10^{-4}

XRD scan of the Fe-6.8Al samples in the three conditions is shown in inset in Fig. 7a confirming the formation of ferrite phase. This is in corroboration to the results published by the same authors in Ref. [15]. Stabilization of the ferrite phase has been reported by the addition of Al to Fe by Rana et al. [3]. Rietveld refinement has been performed for the XRD profiles (inset Fig. 7a) using MAUD software [38] and the results presented in Table 3. For the determination of instrumental broadening, fitting of the XRD profile of Si standard sample was undertaken. It can be observed that the lattice micro-strain decreases after annealing from the cold rolled state which is due to recrystallization of the samples (Table 3). The decrease is sharp from the 98%CR sample to 750 °C, 5 min annealed sample whereas it is shallow from 5 min to 60 min annealed sample as recrystallization is already completed upon annealing at for 5 min [15]. The (110) peak profile of the 98%CR sample shows a large amount of broadening as compared to that for the annealed samples which may be due to the presence of lattice micro-strain imparted during cold rolling. The broadening in XRD peak profile has been described in Ref. [39] to be caused by the presence of lattice imperfections. The two primary lattice imperfections are (i) crystallite size and (ii) lattice strain. The later manifests itself in the form of dislocations. Broadening due to dislocations becomes significant when dislocation density increases above $5\times10^{12}\ m^{-2}$. Fig. 7b–d shows the plots of second order (M_2) and fourth order moment (M_4/q^2), which were obtained after fitting (110) peak with Eq. (7) and Eq. (8) respectively. The dislocation density has been estimated and has been described in Fig. 7e. Fig. 7e shows that the evolution of dislocation density for the three sample conditions. It can be observed that dislocation density decreases rapidly showing a three order drop in magnitude upon annealing till 60 min at 750 °C as compared to the cold rolled sample. This decrease is due to the recrystallization of ferrite upon annealing at 750 °C for 5 min as reported in Ref. [15]. Annealing till longer times (= 60 min) leads to ferrite grain growth [15].

Table 4
Variation of yield strength (YS), ultimate tensile strength (UTS) and uniform elongation of Fe-6.8Al samples in different conditions.

Sample condition	Yield strength (MPa)	Ultimate tensile strength (MPa)	% uniform elongation	Yield ratio (= YS/ UTS)
98%CR	1800	–	1.2	–
750 °C, 5 min	700	870	20.0	0.85
750 °C, 60 min	265	394	29.8	0.67

3.2. Mechanical properties

Fig. 8a shows the engineering stress-strain curves for the Fe-6.8Al low-density steel for the different sample conditions with the values shown in Table 4. The yield strength of the 98% CR sample (= 1800 MPa) is much higher than for the sample annealed at 750 °C for 60 min (= 265 MPa) (Fig. 8a, Table 4). This may be due to the presence of lattice defects (notably dislocations) in the 98%CR sample which acts as barriers to further dislocation propagation and thus increases the yield strength. The yield ratio which is the tendency of the steel to work harden has also been calculated in Table 4. The yield ratio of the fully annealed sample (750 °C, 60 min) is lower than that of the 98%CR sample. The presence of a large number of dislocations for the 98%CR sample inhibits the strain hardening ability leading to higher values of yield ratio. The drastic change in yield strength (1800 MPa, 98% CR to 265 MPa, annealed at 750 °C, 60 min) upon the change in dislocation density from $\sim2.2\times10^{15}\ m^{-2}$ (98%CR) to $\sim10^{12}\ m^{-2}$ (annealed at 750 °C, 60 min) identifies strengthening from dislocations to be one of the main strengthening mechanism. Fig. 8b shows the rate of strain hardening of the Fe-6.8Al annealed at 750 °C for 5 and 60 min. It can be observed that rate of strain hardening for the sample annealed for 60 min to be higher than 5 min. There is a sudden increase in strain hardening rate followed by gradual decrease with further straining.

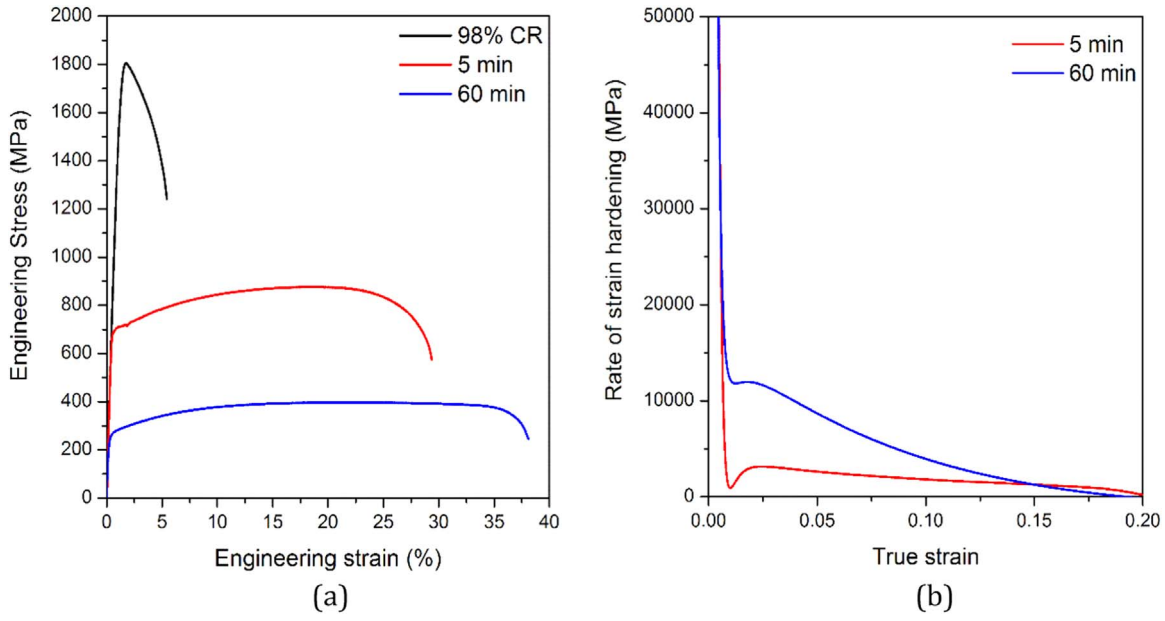


Fig. 8. (a) Engineering stress-strain and (b) rate of strain hardening curves for the Fe-6.8Al low-density steel after 98% cold rolling (CR) and isothermally annealing at 750 °C for 5 and 60 min.

4. Discussion

4.1. Analysis of strengthening mechanisms

To get a better insight into the yield strength of Fe-6.8Al alloy for the cold rolled and annealed samples, the observed yield strength was deconstructed into the contribution of different strengthening mechanisms by the application of empirical formulations. The application of such a procedure in literature has been reported by Kumar et al. [40] to model the strength contribution in Al-Mg-Sc alloy, by Hajy Akbary et al. [41] in Fe-3.5Mn-2.9Si-0.3 C steel and Takaki et al. [42] for a Fe-18Ni-0.001 C steel. In the present study five strengthening mechanisms from (i) dislocations, (ii) grain size, (iii) ordered phase, (iv) solid solution and (v) lattice frictional stress have been used to examine the yield strength of Fe-6.8Al low-density steel in different conditions. Kumar et al. [40] also used the same strengthening mechanisms to correlate the yield strength of Al-Mg-Sc alloy to the various microstructural parameters such as grain size, dislocation density and precipitate size.

The strength contribution from solid solution (τ_{ss}) has been calculated by the application of equation [43]

$$\tau_{ss} = \frac{G\varepsilon_s^{3/2}c^{1/2}}{700},$$
$$\varepsilon_s = |\varepsilon'_G - \beta\varepsilon_b|, \varepsilon'_G = \frac{\varepsilon_G}{(1+\frac{1}{2}|\varepsilon_G|)}, \varepsilon_b = \frac{1}{a}\frac{da}{dc} \text{ and } \varepsilon_G = \frac{1}{G}\frac{dG}{dc} \quad (9)$$

where G is the shear modulus, c is concentration of the solute, a is the lattice parameter, ε_G is the change in shear modulus per unit change in concentration of solute and ε_b is the change in lattice parameter with change in concentration. By applying the assumptions reported in Ref. [43] that: the shear modulus increases linearly with composition, the shear stress required for solid solution strengthening has been derived. For the above calculations the value of G was taken as 82 GPa, concentration (c) was taken as 6.8 wt% and ε_s was taken as 0.45 [43]. The value of τ_{ss} was calculated as 95.6 MPa for the Fe-6.8Al alloy. Using the lattice parameter values measured by XRD in Ref. [15] for Fe-Al alloys, the solid solution strengthening for Fe-Al alloys has been found to increase from 67.8 to 127.2 MPa for the 5 wt% Al to 9 wt% Al respectively. It is to be mentioned here that solid solution strengthening contribution from Fe-Al matrix is only taken into account while doing the above calculation by taking the volume fraction of Fe-Al phase. For very fine domains, it is difficult to separate solid solution strengthening from ordered phase strengthening, hence in such a case the contribution of solid solution strengthening is ignored.

The lattice frictional stress (τ_{PN}) necessary to overcome the interatomic bonding forces during the glide of dislocations on the slip plane was calculated [44] by equation

$$\tau_{PN} = \frac{2G}{(1-\upsilon)}\exp\left(-\frac{2\pi a}{b(1-\upsilon)}\right) \quad (10)$$

where G is the shear modulus, υ is the Poisson's ratio, a is the interplanar spacing and b is the interatomic distance in the slip direction. For the Fe-Al low-density steels system the value of these constants are G = 82 GPa, υ = 0.33, a = 2.02 Å, b = 2.48 Å [43,45]. By substituting the values in Eq. (9), we obtain a value of τ_{PN} ~154.34 MPa. Eq. (10) provides the lattice frictional stress at 0 K [40]. However, due to the presence of thermal energy, the stress value required to move dislocations in the lattice at room temperature would be significantly lower than the value calculated from Eq. (10) [40]. Due to unavailability of lattice frictional stress data in literature pertaining to Fe-Al alloys, we have used a value of 41 MPa which has been used by Speich and Swann [46] for a 0.4 wt% C-steel.

The strengthening due to formation of ordered phase (τ_{order}) can be calculated by the equation

$$\tau_{order} = 0.7G\left(\frac{\gamma}{Gb}\right)^{3/2}\left(\frac{fr}{b}\right)^{1/2} \quad (11)$$

where γ is the antiphase boundary energy, G (= 82 GPa) is the shear modulus, b is the burgers vector, f and r is the fraction and the radius of the ordered phase. The determination of the ordered phase size was done by TEM in Ref. [40] in contrast to both TEM and SAXS based estimation in our present study.

High angle grain boundaries are known to arrest the motion of dislocations [47,48]. The contribution of hardening due to the presence of high angle grain boundaries ($\Delta\sigma_{GB}$)is given by the equation [40]

$$\Delta\sigma_{GB} = k_{HP}D^{-1/2} \quad (12)$$

where D is grain diameter and k_{HP} is a constant. The constant k_{HP} is dependent on the composition of the alloy and its microstructural state. For the present study the Hall-Petch constant for 99.95% pure iron, described in Refs. [49,50], was used to calculate the contribution of grain size strengthening to the overall strength. EBSD scans were done to determine the grain sizes of Al-Mg-Sc alloy in the Ref. [40] whereas in the present study the determination of ferrite grain/crystallite size was done by using SEM images for the annealed samples in conjunction by XRD analysis of the cold rolled samples.

The strengthening contribution ($\Delta\sigma_\rho$) from dislocations has been considered by using the Taylor equation

$$\Delta\sigma_\rho = M\alpha Gb\rho^{1/2} \quad (13)$$

where ρ is the dislocation density, M is the Taylor factor, b is the Burgers vector, G (= 82 GPa) is the shear modulus, α is a constant whose value is usually taken as 0.33 [50]. The dislocation density was calculated by the application of variance method developed by Groma [31] which is applied to (110) XRD peak profile as described in Section 2.3. Whereas the dislocation density in Ref. [40] was determined by TEM where the area under investigation is limited. As compared to the previous studies the present study utilizes XRD to determine dislocation density which takes into account large area for dislocation density analysis. This is in line with Hajy Akbary et al. [41], who utilised XRD analysis (modified Williamson hall and Warren Averbach method) to obtain the dislocation density.

Fig. 9 shows the contribution from each strengthening mechanism to the yield strength of Fe-6.8Al alloy in 98%CR and annealed condition (750 °C: 5 min, 60 min). In the present study, the strengthening contribution from (i) solid solution, (ii) presence of ordered phase and (iii)

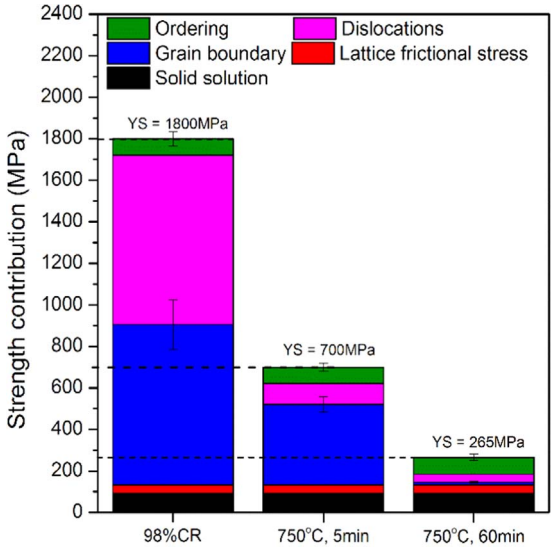


Fig. 9. bar graph showing the contribution of various strengthening mechanisms to the yield strength of Fe-6.8Al low-density steel after (a) 98% cold rolling (CR) and isothermally annealing at 750 °C for (b) 5 min, (c) 60 min.

lattice frictional stress can be assumed to remain constant as they depend upon alloy composition. Error bars can be associated with the strengthening mechanisms of (i) grain boundary and (ii) ordered phase formation. It is due to the incorporation of errors in the determination of ferrite grain size, ordered phase size and volume fraction. For the 98%CR sample, the strengthening from (i) dislocations (generated during rolling) and (ii) grain boundaries are the two dominant mechanisms. But for the annealed samples the strengthening contribution from dislocations decreases due to the reduction in dislocation density and coarsening of ferrite grains upon recrystallization. For the sample annealed for 60 min, the contribution from (i) ordering and (ii) solid solution strengthening are the major contributing mechanisms.

4.2. Strengthening from the ordered phase

It is well established that in polycrystalline materials, strengthening from (i) grain size and (ii) dislocations are the two potential strengthening mechanisms. In the present case, the presence of fine Fe_3Al ordered phase has been established by Mössbauer spectroscopy, SAXS and TEM based studies. Based on the TEM and SAXS results, the ordered phase have been described to be of nano-sized dimensions with sizes ranging between 2.5 and 3.75 nm (Fig. 4). Therefore, the strengthening from nano-sized ordered phase can also be considered to be additional strengthening mechanism operating in the present case. Strengthening from ordered phase formation can be attributed to the difference in separation of the partial dislocations in ordered and disordered phase. This also results in the creation of an antiphase boundary between the two partial dislocations present in the ordered phase. The antiphase boundary is bounded by two dislocations (i) leading dislocation and (ii) trailing dislocation. The leading dislocation destroys the ordered structure whereas the trailing dislocation restores the ordered arrangement. This leads to the constraint on dislocations travelling in pairs thus restricting their mobility and leading to strengthening. In summary, the splitting of dislocations into super-partial dislocation in ordered phases leads to increase in flow stress.

Similar strengthening due to formation of fine particles has also been demonstrated [10,11] in Fe-Mn-Al based low-density steels. Precipitation of nanosized cuboidal κ-carbides having $L'1_2$ type structure have been described by Gutierrez-Urrutia et al. [10] in the Fe-30.5Mn-8.0Al-1.2 C alloy. The shearing of these carbides have been shown to form narrow channels in the austenite matrix leading to significantly improved yield strength. Formation of κ-carbides by spinodal decomposition have been shown in Fe-9Al-28Mn-1.8 C by Chang et al. [11] during quenching of the steel. These carbides provide exceptional strength (YS = 1423 MPa) during tensile testing. However, in the present study the formation of ordered phase has been achieved without major alloying additions.

The yield strength of low-density steel with the composition Fe-6.8Al, annealed at 900 °C for 1 min has been described by Rana et al. [3] to be 342 MPa, respectively. The reported value by Rana et al. [3] is found to be lower than that obtained (YS = 700 MPa) during the annealing of Fe-6.8Al steel at 750 °C for 5 min. This may be due to lower ferrite grain size (= 25.5 ± 5 μm, Ref. [15]) obtained upon annealing in the present case as compared to 69.2 ± 34.8 μm in Ref. [3]. Upon annealing till 60 min at 750 °C the yield strength decreases to 265 MPa which is less than that obtained in Ref. [3] which may be due to higher ferrite grain size. The strong dependence of yield strength on ferrite grain size, proves the existence of Hall-Petch strengthening as one of the main strengthening mechanisms. In the present investigation, low-density steels withAl = 5, 6, 6.8, 9 wt% shows the stabilization of the single ferrite phase upon Al addition. In a few recent publications [8,9], the stabilization of γ along with ferrite has also been shown upon the modification of the composition by the slight addition of Mn (< 4 wt%) and C (< 0.2 wt%) along with Al addition. Although the present investigation reports the formation of single phase, the achieved strength (YS = 700 MPa, elongation = 20%) just after annealing at 750 °C for 5 min rivals to that for the dual phase low-density steels [8]. This could be due to primarily from grain boundary strengthening along with additional contribution from (i) dislocations and (ii) the formation of nano-sized Fe_3Al ordered phase having DO_3 structure. Nevertheless, the composition reported by Rana et al. [8,9] also presents an important milestone for low alloyed low-density steels where the stabilization of duplex phase microstructure consisting of majority ferrite and minor austenite phase fraction was attained for the first time by the addition of relatively very less amount of Mn and C.

5. Conclusions

The present work reports the observation of enhanced yield strength in Fe-Al based low-density steels. One of the reasons for the enhanced strength is due to the formation of nano-sized Fe_3Al ordered phase having DO_3 structure, although the major contributions in the overall strength is due to dislocations and grain size. The characterization of the ordered phase was done by TEM, SAXS and Mössbauer spectroscopic studies. The yield strength of the cold rolled and annealed samples were then evaluated by the numerical formulation of the known strengthening mechanisms. Based upon experimental results and analysis the following conclusions can be drawn.

1. The formation of Fe_3Al ordered phase has been observed in binary Fe-Al based low- density steels. These phases have been characterized by SAXS, Mössbauer spectroscopy and TEM which shows that size of the ordered phase to be between 2.5–3.8 nm. The size of the ordered phase is observed to be increasing with the aluminium content.
2. The tensile yield strengths of the Fe-6.8Al low-density steel samples subjected to different annealing conditions were modelled with the available theoretical formulations for various strengthening mechanisms. For the cold rolled sample, strengthening contributions from (i) dislocations (generated during cold rolling) and (ii) grain size/boundaries were observed to be the dominant. For the fully recrystallised samples, the contribution to the strengthening mechanisms was in the following order: (i) grain size strengthening (dominant mechanism), followed by nearly equal contribution from (ii) dislocations and (iii) ordered Fe_3Al phase. The material annealed even beyond complete recrystallization underwent higher extent of ordering, and hence overall strength was due to nearly equal contributions from (i) solid solution strengthening and (ii) order strengthening.

In summary, it can be stated that the dominance and interplay of various strengthening mechanisms including order strengthening leads to the enhanced strengthening in Fe-Al alloys.

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